
Text Classification with Deep Learning: Character-Level Recurrent Networks vs. Parameter-Efficient Transformers

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Abstract

We study binary sentiment classification on a Chinese review corpus, comparing a character-level bidirectional LSTM with a pretrained RoBERTa-based classifier. For the advanced task we extend the RoBERTa baseline with layer mixing, gated multi-view pooling, and parameter-efficient adaptation via LoRA and BitFit. Under a controlled matched-baseline protocol the advanced architecture improves validation macro F1 from 0.9085 to 0.9317 (+0.0232), reduces validation loss from 0.1711 to 0.1435, and achieves 95.80% accuracy. The strengthened frozen baseline already uses gated multi-view pooling, so the remaining gain isolates the value of layer mixing and parameter-efficient encoder adaptation on top of that stronger head.

1 Introduction

Sentiment analysis over Chinese-language text presents a practical and well-scoped setting for evaluating neural text classifiers. Unlike English, Chinese lacks whitespace-delimited tokens, which makes the choice of segmentation granularity—character, word, or subword—a consequential design decision. Moreover, the prevalence of short, colloquial reviews combined with a moderate degree of class imbalance demands that classifiers generalise effectively beyond the dominant class.

This project addresses binary sentiment classification on a Chinese review dataset and pursues two complementary goals. The *basic task* compares two substantially different architectural families: (i) a lightweight character-level bidirectional LSTM trained entirely from scratch, and (ii) a pretrained Chinese RoBERTa encoder fine-tuned with a simple pooled classification head. The *advanced task* then develops a stronger system by modifying the RoBERTa-based model through a combination of multi-view sentence-representation learning and parameter-efficient adaptation.

The main contributions of the report are as follows. First, we provide a rigorous empirical comparison of recurrent and Transformer architectures on the same dataset under identical evaluation conditions, demonstrating that pretrained contextual representations provide a substantial advantage. Second, we introduce and evaluate a controlled architectural modification that adds layer mixing, max-augmented pooling, and LoRA-based adaptation on top of a stronger frozen RoBERTa baseline, showing a macro F1 improvement of +0.0232 over the matched baseline. Third, we show that the improvement survives migration from a hyperparameter-search sandbox into the project training pipeline, confirming reproducibility. Finally, we present a detailed analysis connecting the observed performance gains to specific architectural choices, informing future design decisions.

2 Dataset and Preprocessing

2.1 Dataset

The dataset consists of Chinese consumer reviews labelled for binary sentiment. Each example contains a single review text field and a binary label field. The training set contains 10,000 examples; the validation set contains 2,500. Table 1 summarises the key statistics.

Table 1: Dataset statistics.

Split	Samples	Positive	Negative	Pos. Ratio	Avg. Chars
Train	10,000	8,066	1,934	80.66%	53.29
Validation	2,500	2,025	475	81.00%	54.69

Two properties of the data drive the experimental design. First, the class distribution is moderately imbalanced: positive examples account for approximately 81% of both splits. A classifier that predicts the majority class always would achieve ~81% accuracy yet zero minority-class recall, making accuracy an insufficient primary metric. We therefore adopt *macro F1* as the principal model-selection criterion throughout. Second, the average review length is only 53–55 characters, but a small number of examples exceed 1,000 characters, necessitating explicit sequence truncation.

2.2 Preprocessing Pipeline

Character-level preprocessing (BiLSTM). The LSTM model operates directly on the raw character sequence. A character vocabulary is constructed from the training set with a minimum frequency threshold of 2; characters appearing fewer than twice are mapped to a shared <unk> token. The resulting vocabulary size is 3,027 including the <pad> and <unk> special tokens. All sequences are truncated or padded to a maximum length of 512 characters. No lowercasing or normalisation beyond <unk> substitution is applied.

Subword preprocessing (BERT models). Both RoBERTa-based models use the Hugging Face tokenizer shipped with `hfl/chinese-roberta-wwm-ext` [Cui et al., 2020]. This tokenizer uses whole-word masking-aware subword vocabularies of size 21,128 and handles Chinese text directly without external word segmentation. All sequences are truncated or padded to a maximum length of 256 tokens, which is sufficient to cover the vast majority of validation examples without loss.

3 Model Architectures

3.1 Basic Model 1: Character-Level BiLSTM

The BiLSTM classifier treats each review as a sequence of character indices and processes it as follows.

1. **Embedding layer.** Each character index is mapped to a 64-dimensional trainable embedding vector.
2. **Encoder.** A single-layer bidirectional LSTM with hidden size 64 (total state dimension 128) encodes the sequence.
3. **Pooling.** The final forward and backward hidden states are concatenated into a 128-dimensional sentence vector.
4. **Regularisation.** Dropout with probability 0.3 is applied to the pooled vector.
5. **Classification head.** A linear layer maps the 128-dimensional vector to a 2-class logit.

The model has 260,546 total (and trainable) parameters. Its advantage is simplicity and fast training, but it cannot exploit pretrained linguistic knowledge and must learn all sentiment-relevant representations from supervised signal alone.

3.2 Basic Model 2: Baseline RoBERTa Classifier

The second model uses a pretrained Chinese RoBERTa encoder [Cui et al., 2020] as a fixed feature extractor with a stronger trainable pooling head.

1. **Encoder.** hfl/chinese-roberta-wwm-ext, a 12-layer Transformer with 768-dimensional hidden states and 12 attention heads.
2. **Pooling.** The final-layer token embeddings are summarized with four fused views: the configured anchor view ([CLS] in our run), masked mean pooling, learned attention pooling, and a gated combination of those three views. The concatenated representation is layer-normalized before classification.
3. **Projection MLP.** A linear projection from the 3072-dimensional fused representation to a 512-dimensional hidden representation, followed by GELU activation and dropout with probability 0.08.
4. **Classification head.** A linear layer producing 2-class logits.

The encoder is fully frozen; only the pooling and classification head are trained, giving 1,588,230 trainable parameters out of 103,855,878 total. Using a frozen encoder still keeps optimization cheap and stable, but the richer pooling head makes much better use of the pretrained sentence features than a raw [CLS] projection alone.

3.3 Advanced Task: Baseline and Modified Architecture

Controlled frozen baseline. For the advanced task we construct a *matched frozen baseline* that shares the same backbone, optimiser schedule, hidden dimension (512), dropout (0.08), batch size (12), and random seed as the advanced model, but (i) freezes all encoder parameters and (ii) uses final-layer pooling only, without layer mixing, max pooling, or encoder adaptation. The baseline pools the frozen token embeddings through [CLS], masked mean, learned attention, and a gated fusion of those views. This baseline has 1,588,230 trainable parameters.

Advanced BERT architecture. The advanced classifier extends the baseline in five directions, each targeting a known limitation of the frozen [CLS]-head design.

1. **Layer mixing.** Rather than consuming only the final encoder layer, the model learns a softmax-normalised scalar mixture of the top 4 hidden layers. Concatenating representations across abstraction levels allows the classifier to leverage both fine-grained surface features (captured at lower layers) and coarser semantic compositionality (at higher layers) [Tenney et al., 2019].
2. **Multi-view pooling.** Four complementary sentence views are computed over the mixed representations:
 - [CLS] token: a global sequence summary distilled during pretraining;
 - masked mean pooling: stable average representation;
 - learned attention pooling: task-driven weighted sum;
 - masked max pooling: captures the most salient token per dimension, useful for polarity-anchoring content words. Relative to the frozen baseline, this adds the max-pooled channel on top of an already strong [CLS] + mean + attention fusion head.
3. **Gated fusion.** A learned gating network weights the four pooled views per example. The gated weighted sum is concatenated with the original views and layer-normalised before classification, giving the model input-adaptive sentence representation.
4. **Parameter-efficient adaptation.** LoRA adapters [Hu et al., 2022] are injected into the query, key, and value projections of all 12 encoder layers (rank 24, $\alpha = 32$), and into the attention output projection of the top 4 layers. BitFit [Ben Zaken et al., 2022] additionally enables all encoder bias parameters. LayerNorm parameters remain frozen. This allows the backbone to adapt to the sentiment domain while keeping the vast majority of parameters frozen.
5. **Wider classification head.** The fused representation is projected to a 512-dimensional hidden layer via GELU and dropout 0.08, increasing head capacity to match the richer input.

The advanced model has 105,730,571 total parameters but only 3,565,835 trainable parameters (3.37% of total), making it more parameter-efficient than the full fine-tuning alternative.

Table 2 summarises the architectural differences.

Table 2: Architecture comparison: controlled frozen baseline vs. advanced BERT.

Component	Frozen Baseline	Advanced BERT
Encoder usage	Final layer only, frozen	Learned mix of top-4 layers
Pooling	[CLS] + mean + attn + gated fusion	[CLS] + mean + attn + max + gated fusion
Adaptation	None	LoRA (rank 24) + BitFit
Classifier head	512-dim MLP on fused representations	512-dim MLP on richer fused representations
Trainable params	1,588,230	3,565,835

4 Training Setup

4.1 Common Implementation Details

All models are trained using the shared pipeline in `train.py` with PyTorch. The following choices apply uniformly: cross-entropy loss, AdamW optimiser [Loshchilov and Hutter, 2019], checkpoint selection by maximum validation macro F1, TensorBoard logging under `runs/`, and fixed random seed 42. The best checkpoint for each experiment is saved to `checkpoints/<experiment_name>/best_model.pt`.

4.2 Per-Model Hyperparameters

Table 3 lists the key hyperparameters for each experiment. Crucially, the controlled frozen baseline and the advanced BERT model share all non-architectural settings, so any observed difference in validation performance is attributable to the architectural changes described in Section 3.3.

Table 3: Training hyperparameters. The controlled baseline and advanced BERT share all settings except architectural ones (marked with †).

Hyperparameter	BiLSTM	Frozen Baseline	Advanced BERT
Optimizer	AdamW	AdamW	AdamW
Learning rate (head/main)	10^{-3}	5.13×10^{-5}	5.13×10^{-5}
Learning rate (encoder)	—	—	3.91×10^{-5}
Weight decay	0.05	0.00169	0.00169
Batch size	64	12	12
Epochs	15	4	4
Warmup ratio	—	0.150	0.150
Final LR scale	—	0.05	0.05
Dropout	0.30	0.08	0.08
Classifier hidden	—	512	512
Max sequence length	512 chars	256 tokens	256 tokens
LoRA rank / α †	—	—	24 / 32
Layer mix depth†	—	—	4
Encoder frozen†	—	Yes	No (LoRA+BitFit)

5 Experiments

5.1 Basic Task Comparison

The basic task pitches the character-level BiLSTM (Section 3.1) directly against the RoBERTa baseline (Section 3.2). The comparison is designed to measure the magnitude of the advantage conferred by large-scale language pretraining under the same dataset and evaluation conditions. Both

models are trained to convergence according to their respective early-stopping criteria (validation macro F1 on held-out data).

5.2 Advanced Task Comparison

For the advanced task we follow a controlled experimental protocol. The frozen RoBERTa baseline is retrained with hyperparameters that exactly mirror those of the advanced model (see Table 3). The only intended differences are architectural: the matched baseline freezes the encoder and uses only the [CLS] token, while the advanced model adds layer mixing, multi-view pooling, gated fusion, and parameter-efficient encoder adaptation. This protocol does not individually ablate each advanced component, but it does cleanly isolate the aggregate effect of the full architectural upgrade from confounds related to optimiser schedule or regularisation.

6 Results

6.1 Main Quantitative Results

Table 4 reports the primary quantitative results. All metrics are measured on the held-out validation set using the best checkpoint of each run (selected by macro F1).

Table 4: Validation-set results. All values come directly from the saved best checkpoint of each run. Macro F1 is the primary selection metric.

Model	Acc.	Prec.	Recall	F1	Macro F1	Val. Loss
BiLSTM (char-level)	0.9252	0.9505	0.9575	0.9540	0.8770	0.2596
Frozen RoBERTa baseline	0.9420	0.9728	0.9551	0.9639	0.9085	0.1711
Advanced BERT	0.9580	0.9738	0.9743	0.9741	0.9317	0.1435

Table 4 reveals three clear findings. First, the BiLSTM achieves a respectable 92.52% accuracy and 0.8770 macro F1, demonstrating that character-level recurrent models can handle short Chinese reviews adequately. Second, the strengthened frozen RoBERTa baseline already outperforms the BiLSTM across all classification metrics, reaching 94.20% accuracy and 0.9085 macro F1 with substantially lower validation loss (0.1711 vs. 0.2596). Third, the advanced BERT still achieves the strongest results across all six metrics, improving on the frozen baseline by 1.60 accuracy points and 0.0232 macro F1 while further reducing validation loss.

6.2 Advanced Task: Gain over Frozen Baseline

Table 5 quantifies the per-metric improvement of the advanced model relative to the matched frozen baseline.

Table 5: Absolute metric improvement of Advanced BERT over the controlled frozen baseline (Δ = Advanced – Baseline; for loss, negative Δ is better).

	Acc.	Prec.	Recall	F1	Macro F1	Val. Loss
Δ	+0.0160	+0.0010	+0.0193	+0.0102	+0.0232	−0.0276

The largest relative improvement occurs on macro F1 (+0.0232), the metric most sensitive to minority-class behaviour. This suggests that the multi-view representation upgrade and encoder adaptation primarily help the model better identify negative sentiment examples—the harder, minority class.

6.3 Training Curves and Learning-Rate Schedules

Figure 1 shows the training and validation curves for the controlled frozen baseline and the advanced BERT. The advanced model reaches a consistently higher macro F1 plateau and lower best-checkpoint loss. The frozen baseline peaks at step 2085 (macro F1 = 0.9085), while the advanced model reaches

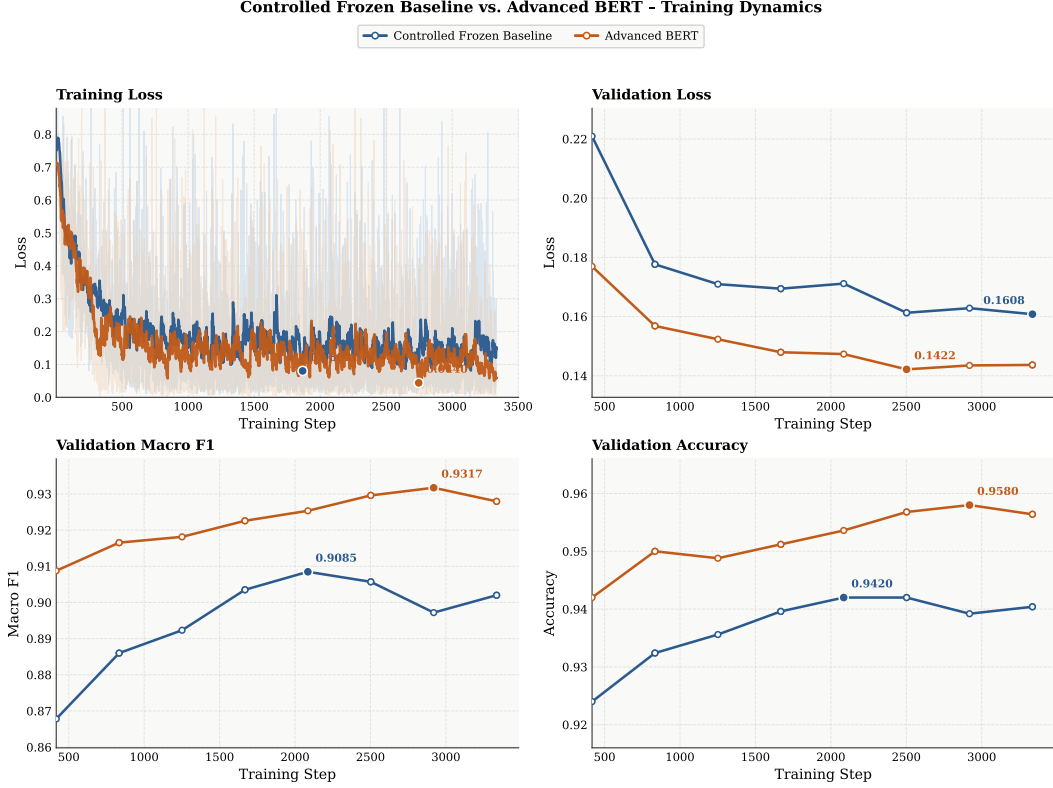


Figure 1: Training dynamics for the controlled frozen baseline (blue) and the advanced BERT (orange). **Top left:** step-level training loss with raw signal (pale) and EMA-smoothed overlay (solid). **Top right:** validation loss at fixed evaluation intervals. **Bottom left:** validation macro F1. **Bottom right:** validation accuracy. Filled circles with annotations mark the best-checkpoint step for each run.

its best checkpoint at step 2919 (macro F1=0.9317), indicating that the richer architecture still extracts additional value beyond the stronger frozen pooling head.

Figure 2 shows the learning-rate schedules logged during training. Both models share the same warmup-plus-cosine-decay shape. The advanced model records a separate, lower LR for the encoder parameter group (green dashed), consistent with its intent to adapt the pretrained backbone gently without catastrophic forgetting.

Figure 3 presents a compact bar-chart comparison of accuracy, macro F1, and validation loss at the best checkpoint of each model. The Δ annotations (green) confirm consistent improvement across all metrics, with the largest relative gain on macro F1 (+0.0232).

Figure 4 contextualises parameter efficiency across all three models. Despite sharing a $\sim 104\text{--}106$ M-parameter backbone, the advanced model activates only 3.57 M trainable parameters via LoRA and BitFit—about $2.2\times$ more than the frozen baseline’s 1.59 M, yet still a small fraction of the cost of full fine-tuning.

7 Discussion

7.1 Comparison of the Basic Models

The most striking result from the basic-task comparison is that the strengthened frozen RoBERTa baseline now *outperforms* the character-level BiLSTM on accuracy, F1, macro F1, and validation loss despite keeping the pretrained encoder entirely fixed. This indicates that the earlier weakness of the frozen Transformer was not due to frozen representations alone, but to an underpowered pooling

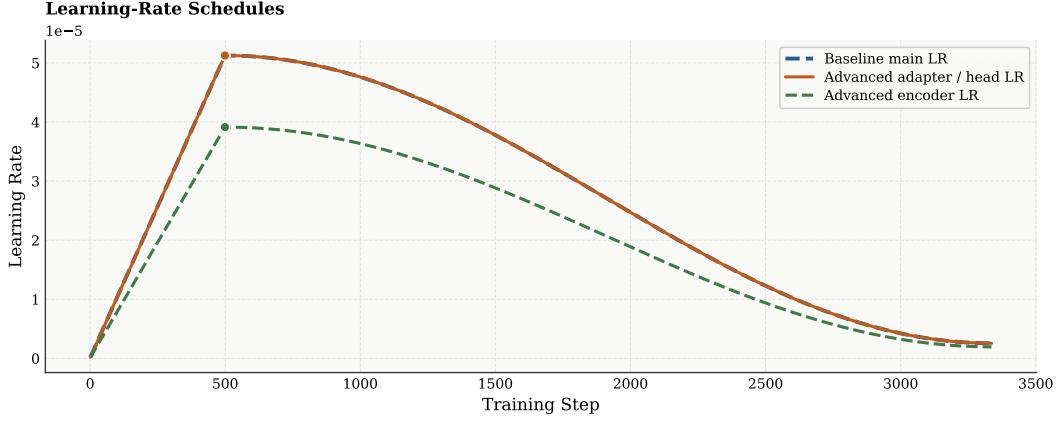


Figure 2: Learning-rate schedules. **Blue dashed:** main LR for the frozen baseline. **Orange solid:** adapter and classification-head LR for the advanced BERT. **Green dashed:** encoder LR for the advanced BERT, held lower to protect pretrained representations. The baseline and advanced adapter schedules are identical, so the dashed baseline trace is intentionally styled to remain visible on top of the solid orange line.

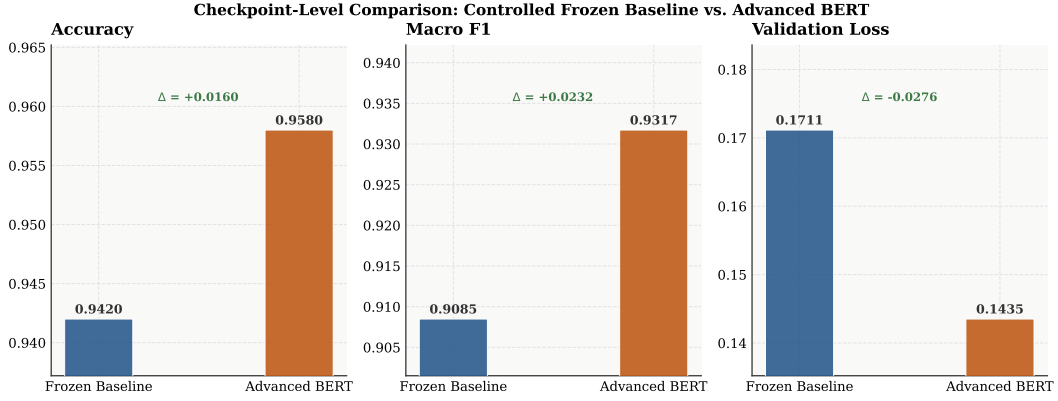


Figure 3: Best-checkpoint comparison of the three headline metrics for the controlled frozen baseline (blue) and the advanced BERT (orange). Higher is better for accuracy and macro F1; lower is better for validation loss. Green Δ labels quantify the direction and magnitude of improvement for the advanced model.

head. Once the baseline aggregates [CLS], mean, and attention-pooled views with a small gate, the pretrained encoder provides a clear advantage over the recurrent model on this dataset.

The BiLSTM remains competitive: it reaches 0.8770 macro F1 with only 260 K parameters and full end-to-end optimization. That makes it a strong lightweight baseline. However, the RoBERTa model benefits from large-scale pretraining and a richer sentence representation, which together outweigh the BiLSTM’s end-to-end trainability on this task.

Regarding convergence behaviour, the BiLSTM is trained for 15 epochs with a $1,000\times$ larger learning rate and converges stably within that budget, whereas the frozen baseline uses far fewer gradient updates (4 epochs) at a smaller learning rate. The architectures are therefore not trained under identical compute budgets, which should be kept in mind when comparing absolute metric values.

7.2 Effect of the Architectural Modification

Under the matched controlled protocol the advanced BERT improves macro F1 by 0.0232 over an already strong frozen baseline. Because that baseline already includes gated [CLS] + mean +

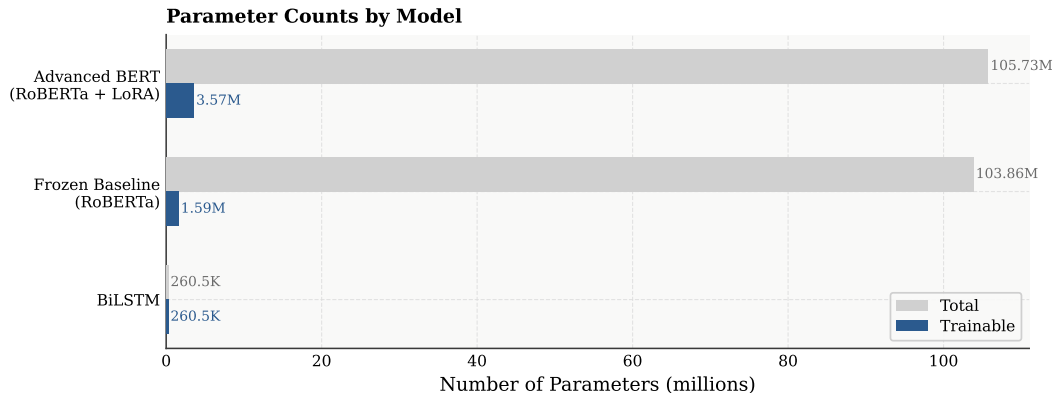


Figure 4: Total (gray) vs. trainable (blue) parameter counts for each model. The BiLSTM is fully trainable but tiny (≈ 261 K). Both BERT-based models share a ~ 104 – 106 M-parameter backbone; the advanced model activates about $2.2\times$ more trainable parameters than the frozen baseline by injecting LoRA adapters and enabling BitFit, while keeping the vast majority of the pretrained encoder frozen.

attention pooling, the remaining gain is more focused and interpretable than before. We attribute it to three complementary mechanisms.

Layer mixing. Different encoder layers encode different linguistic abstractions [Tenney et al., 2019]. By taking a weighted mixture of the top 4 layers the advanced model can simultaneously use shallow syntactic signals and deep semantic compositionality, both of which are relevant for sentiment.

Richer multi-view pooling. The baseline already fuses [CLS], mean, and attention pooling, so the advanced model’s remaining pooling gain comes from adding the max-pooled channel on top of those views. This helps preserve polarity-bearing tokens that can otherwise be diluted by average-based summaries.

LoRA and BitFit adaptation. The pretrained encoder was optimised for a masked-language-modelling objective on a broad web corpus, not for binary sentiment classification on consumer reviews. LoRA and BitFit provide a parameter-efficient channel through which the encoder can adjust to the target distribution without catastrophically forgetting its pretrained representations. The improvement in minority-class recall (negative sentiment) is consistent with this interpretation: domain adaptation helps the model activate features specific to the less frequent class.

It is important to note that the advanced architecture bundles several modifications together, so we cannot attribute the full $+0.0232$ gain to any single component. A full per-component ablation remains future work.

7.3 Limitations

Several methodological limitations affect the strength of the conclusions. First, all experiments are single-seed; variance across random seeds is unknown. Second, the class imbalance (81% positive) is not addressed by any explicit balancing strategy such as class-weighted loss or oversampling, which could further improve minority-class performance. Third, while the advanced task comparison is carefully controlled at the optimiser level, the advanced model was initially designed with the benefit of a preceding hyperparameter search, which may introduce implicit selection bias. Fourth, the evaluation is restricted to one dataset, limiting the generalisability of the conclusions.

8 Conclusion

We presented an empirical evaluation of three sentiment classifiers on a Chinese binary review dataset. The BiLSTM provides a competitive character-level baseline with strong macro F1 (0.8770) under fully supervised training. A strengthened frozen RoBERTa baseline with gated multi-view pooling already reaches 0.9085 macro F1 and 94.20% accuracy, showing that frozen pretrained representations can be highly effective when pooled well. Most importantly, the advanced BERT

classifier—which combines layer mixing, max-augmented pooling, and parameter-efficient LoRA and BitFit adaptation—achieves validation macro F1 of 0.9317 and accuracy of 95.80%, outperforming the matched frozen baseline under identical non-architectural conditions.

The key lesson is that *how* encoder features are aggregated and *where* adaptation is permitted inside a pretrained network both matter substantially. Better frozen-head pooling closes much of the gap, but layer mixing and lightweight encoder adaptation still provide a meaningful final gain.

Future work should systematically ablate each advanced component to quantify individual contributions; evaluate robustness across multiple random seeds; explore class-balanced training to further reduce minority-class error; and test transferability of the advanced architecture to other Chinese sentiment or topic classification datasets.

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A Checkpoint Metadata

The following checkpoint directories contain the model weights used to compute all reported metrics.

- checkpoints/lstm_classifier/best_model.pt
- checkpoints/bert_classifier/best_model.pt
- checkpoints/bert_classifier_advanced/best_model.pt

TensorBoard logs are stored under runs/ and can be visualised with:

```
tensorboard --logdir ./runs
```

The most relevant event files are

- `runs/bert_classifier/bert_classifier_20260331-220647` (older frozen baseline run, retained for reference)
- `runs/bert_classifier/bert_classifier_20260408-222445` (updated frozen baseline, best at step 2085, macro F1 = 0.9085)
- `runs/bert_classifier_advanced/bert_classifier_advanced_20260331-025421` (advanced BERT reproduced run, best at step 2919, macro F1 = 0.9317)

B Implementation Notes

All experiments were run with PyTorch via the uv virtual environment pinned to Python 3.11. Figure generation is handled by `scripts/export_report_figures.py`, which reads TensorBoard event files and writes publication-quality `.pdf` and `.png` exports under `figures/`.